

Camfranglais

How the app works.
The essential source files.

CS4110 / SUMMER 2026 / SET A

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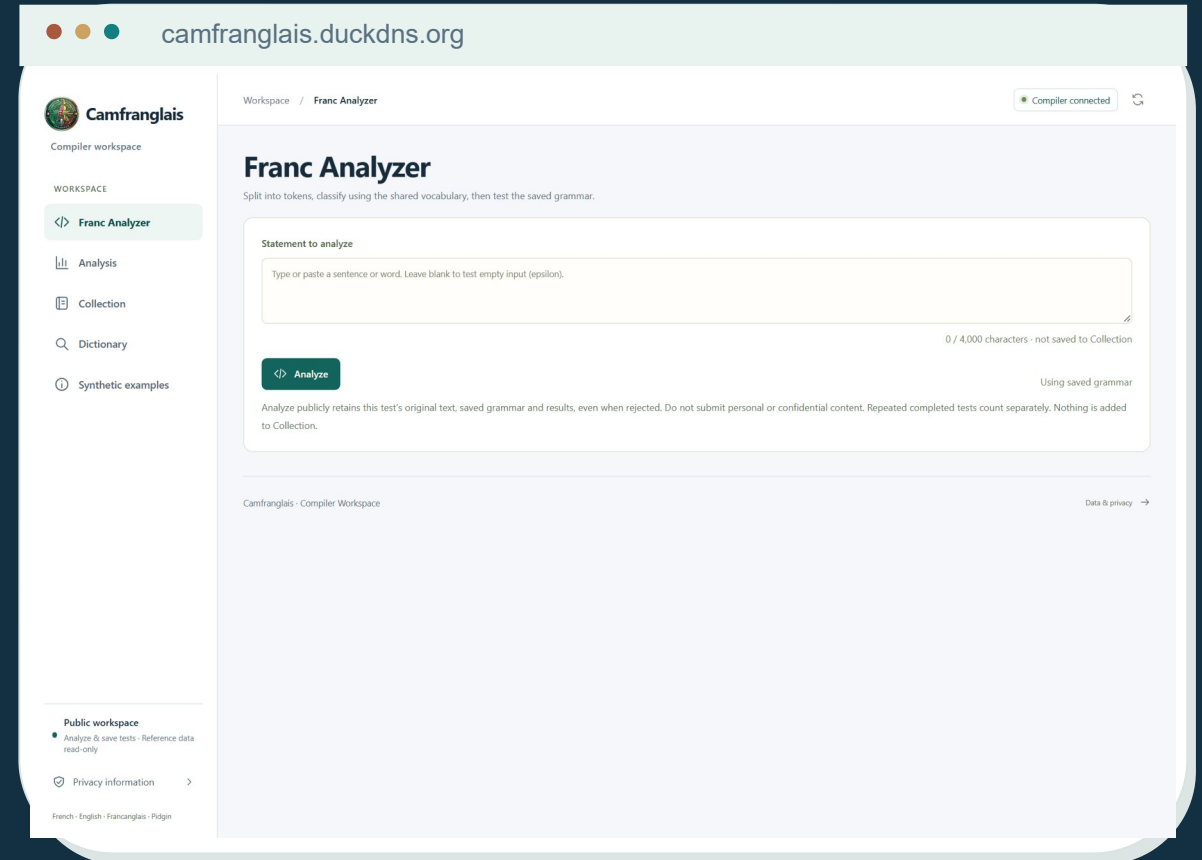
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IMPLEMENTATION / EVIDENCE

README.md

CS4110 / SET A / 27 SEP 2026



docs/document-style.tex

Our goal: make language analysis visible

A transparent teaching tool for Camfranglais statements.

01

PRESERVE

Keep the original words,
spelling and punctuation.

02

EXPLAIN

Show token categories
and the grammar decision.

03

REVIEW

Save the result and
inspect its evidence.

IMPLEMENTATION / EVIDENCE

compiler/lexer/tokenizer.py
backend/analyzer_history.py
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compiler/parser/predictive.py

Five pages, one public workspace

No login: analyze and save tests; reference data stays read-only.

01

Franc Analyzer

Enter a statement and run a test.

02

Analysis

Inspect saved results and statistics.

03

Collection

Browse supplied statements and audio.

04

Dictionary

Look up reference vocabulary.

05

Synthetic examples

Try clearly labelled practice material.

**ONE SHARED
PUBLIC WORKSPACE**

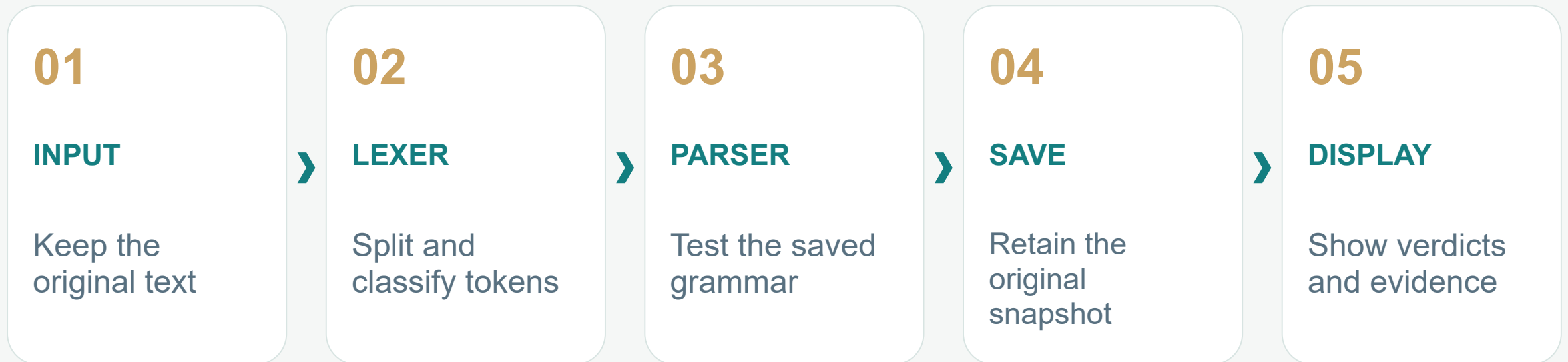
IMPLEMENTATION / EVIDENCE

frontend/src/App.tsx
backend/public_access.py
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frontend/src/PublicSession.tsx

What happens when we press Analyze?

One input becomes a traceable, saved result.



FOLLOW THE DATA / KEEP EACH RESPONSIBILITY VISIBLE

IMPLEMENTATION / EVIDENCE

frontend/src/FrancAnalyzer.tsx
compiler/lexer/tokenizer.py
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backend/analyzer_api.py
compiler/parser/predictive.py

Franc Analyzer: the main action

Type or paste a statement, then explicitly choose Analyze.

Original text

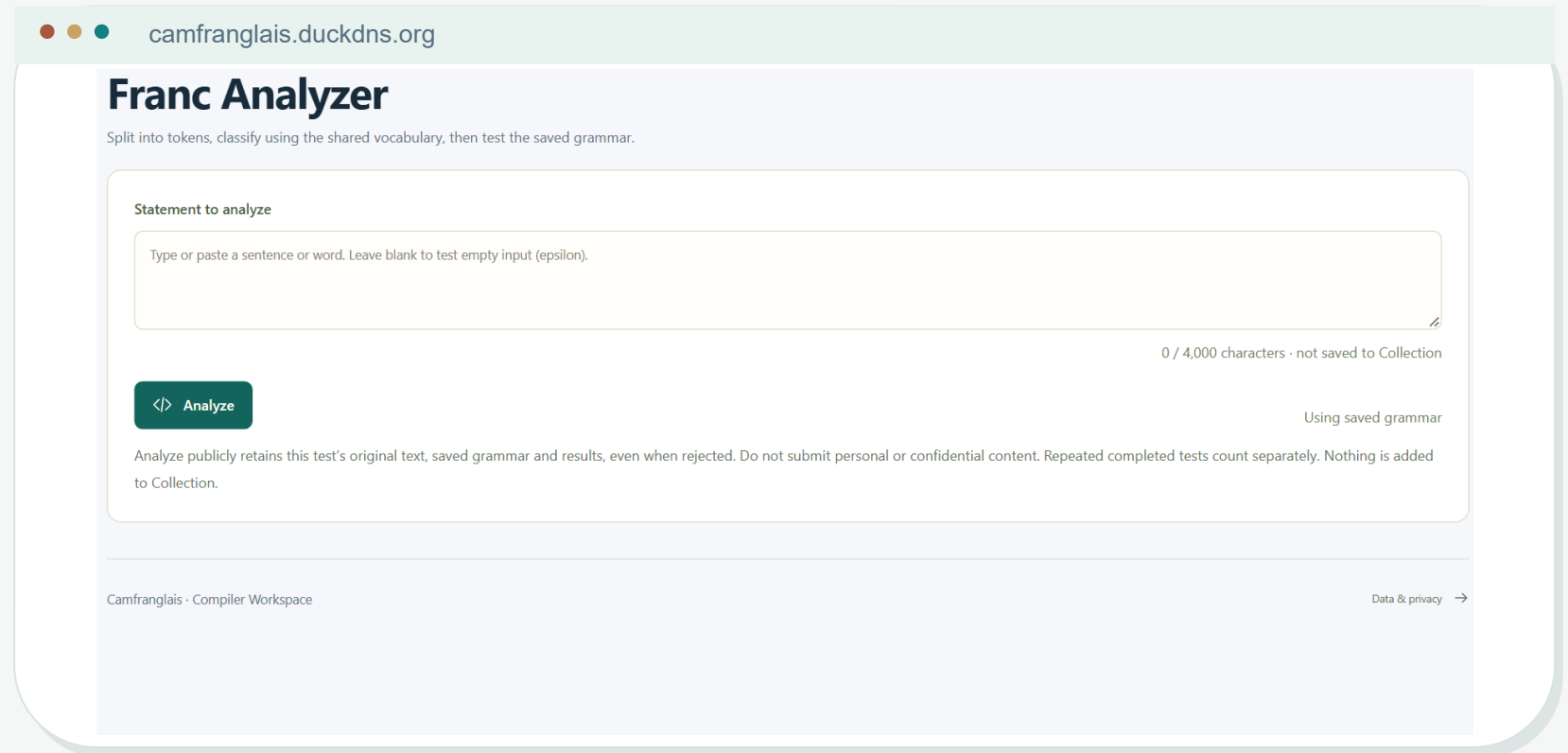
No silent spelling correction.

Saved grammar

The public visitor cannot edit it.

Public result

The test is retained, not added to Collection.



The screenshot shows a web browser window with the address bar displaying 'camfranglais.duckdns.org'. The page title is 'Franc Analyzer'. Below the title, a subtitle reads 'Split into tokens, classify using the shared vocabulary, then test the saved grammar.' The main content area features a text input field with the placeholder text 'Type or paste a sentence or word. Leave blank to test empty input (epsilon)'. To the right of the input field, it shows '0 / 4,000 characters · not saved to Collection'. Below the input field is a green button with a code icon and the text 'Analyze'. At the bottom of the main content area, a disclaimer states: 'Analyze publicly retains this test's original text, saved grammar and results, even when rejected. Do not submit personal or confidential content. Repeated completed tests count separately. Nothing is added to Collection.' The footer of the page includes 'Camfranglais · Compiler Workspace' on the left and a 'Data & privacy' link with a right arrow on the right.

REAL DEPLOYED APP / 27 SEP 2026 / crop only where noted

IMPLEMENTATION / EVIDENCE

frontend/src/FrancAnalyzer.tsx
backend/analyzer_history.py
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backend/analyzer_api.py

Two verdicts explain the result

Example: je suis kass je go nang

Vocabulary

Passes when no token is UNKNOWN.

Grammar

Passes when categories fit the saved rules.

Evidence

S02: six tokens consumed; the CFG accepts.

camfranglais.duckdns.org

SELECTED SAVED TEST

Analyzed sentence or word

Recorded 9/25/2026, 7:49:11 AM. These are the original results, not a new analysis using today's settings.

Publicly retained original text and results. Saved tests are immutable.

je suis kass je go nang

ACCEPT

No UNKNOWN tokens. Recognized slang and all other known lexer categories are accepted.

Vocabulary approval is not grammatical correctness. The CFG grammar check is shown separately in Analysis.

Use as analyzer input →

▼ Token details for this test

#	Observed text	Lexer category
1	je	FRENCH_FUNCTION_WORD
2	suis	VERB
3	kass	ADJECTIVE
4	je	FRENCH_FUNCTION_WORD
5	go	VERB
6	nang	VERB

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IMPLEMENTATION / EVIDENCE

frontend/src/CourseworkResults.tsx
compiler/parser/predictive.py
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frontend/src/VocabularyVerdict.tsx

Reference pages support the analysis

Browse the evidence without changing it.

Collection

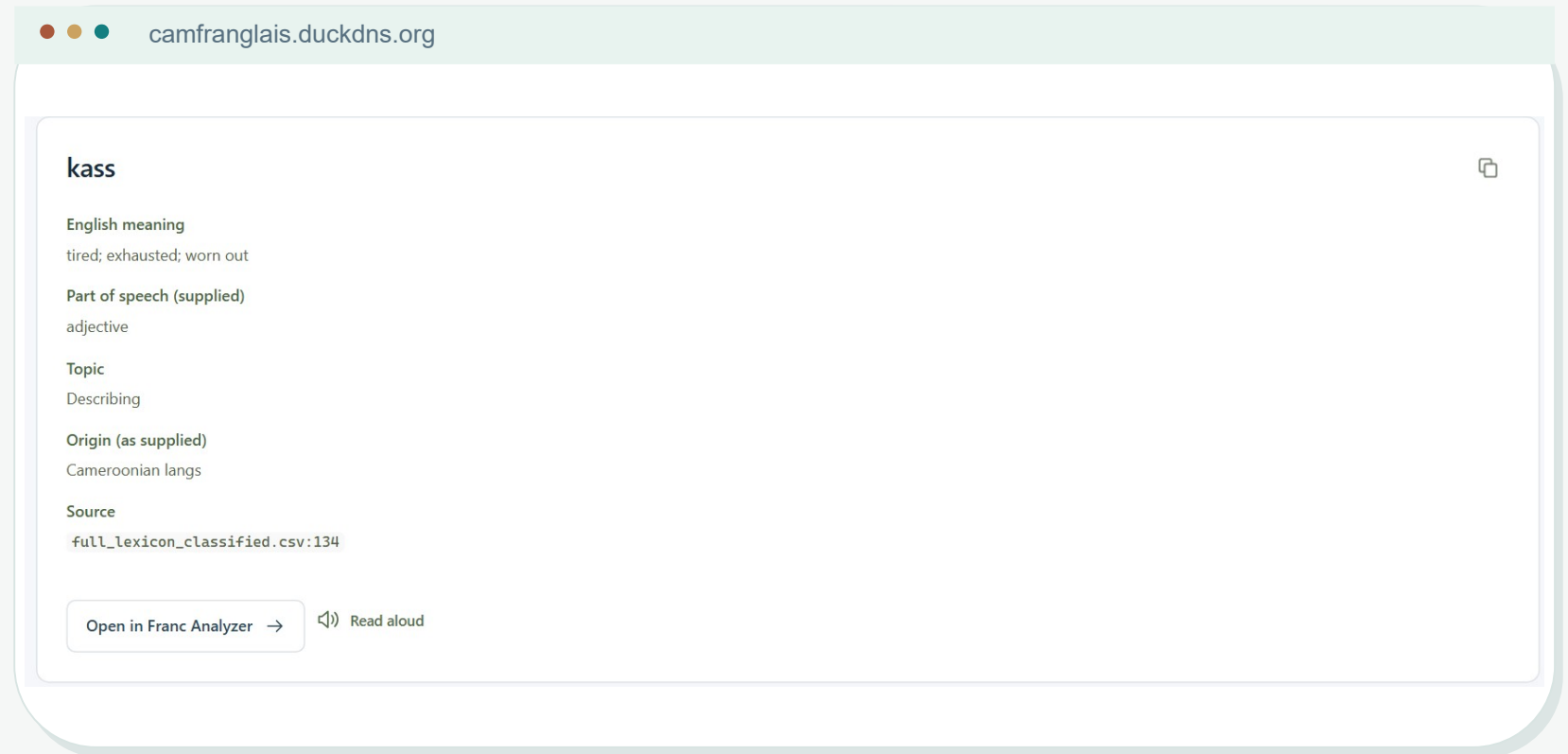
Original statements, context and existing audio.

Dictionary

938 source-labelled reference entries.

Examples

26 constructed practice statements, not fieldwork.



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IMPLEMENTATION / EVIDENCE

frontend/src/Collection.tsx
frontend/src/Examples.tsx
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frontend/src/Dictionary.tsx
backend/dictionary.py

Three layers and their key files

Each layer has one clear responsibility.

React interface

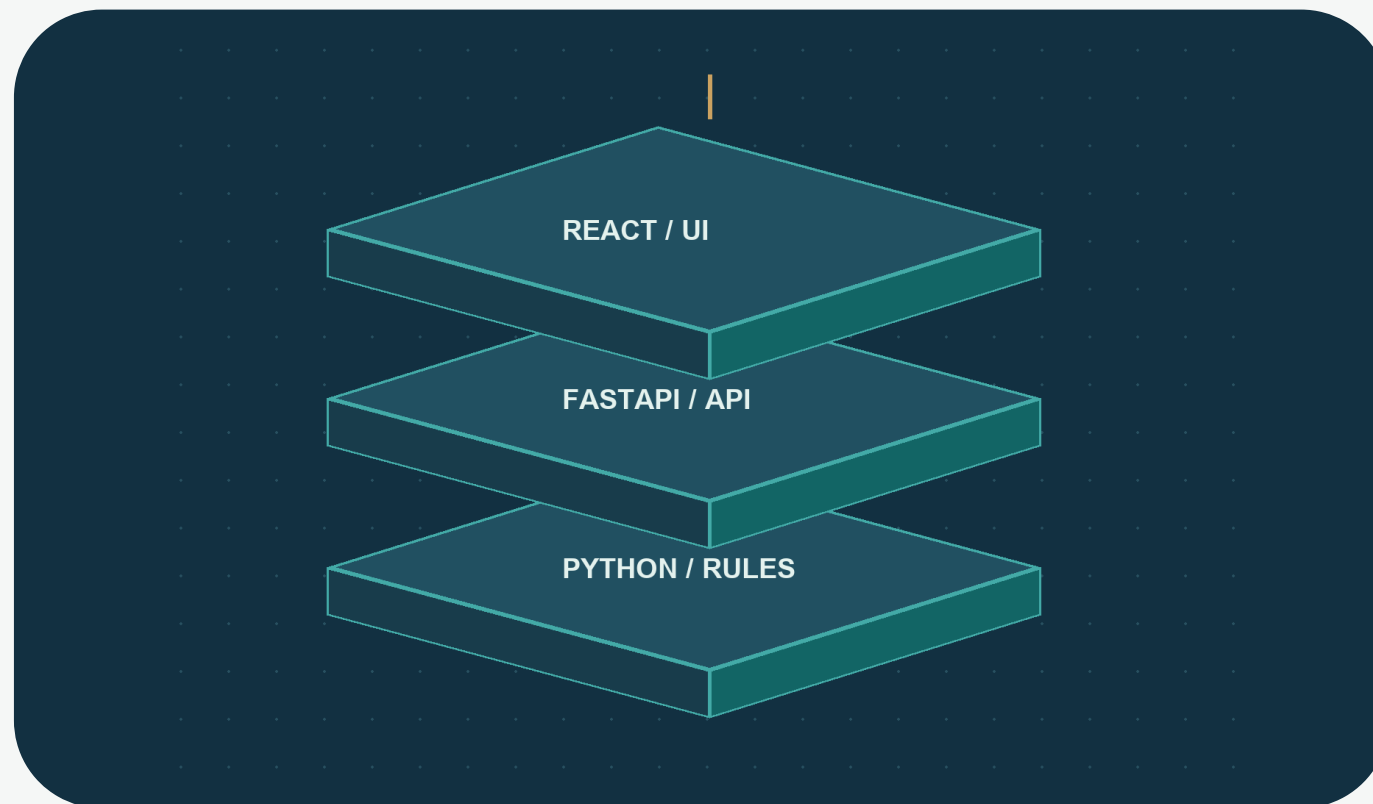
App.tsx: pages and navigation.

FastAPI backend

main.py: public API wiring.

Python compiler

tokenizer.py and predictive.py: the analysis rules.



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frontend/src/App.tsx
compiler/lexer/tokenizer.py
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backend/main.py
compiler/parser/predictive.py

Lexer: from characters to tokens

An equivalent DFA explains the actual segmentation regex.

Recognize

Word, number and punctuation forms.

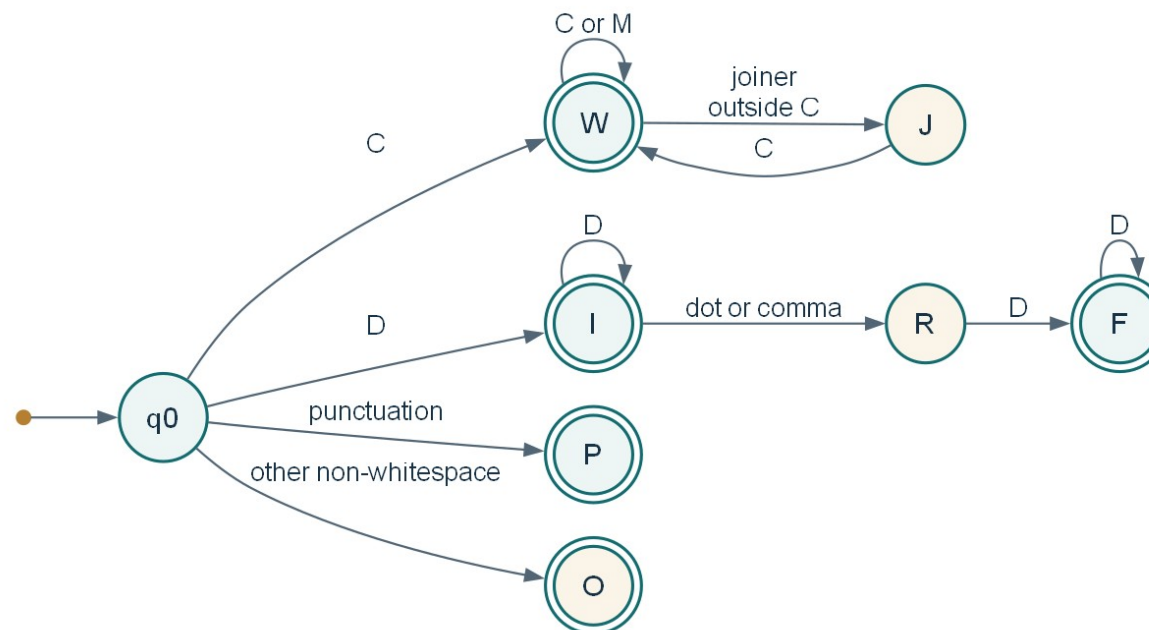
Classify

Apply ordered vocabulary and regex rules.

Preserve

Unknown characters are kept, not discarded.

LEXER / FINITE-STATE TOKEN RECOGNITION



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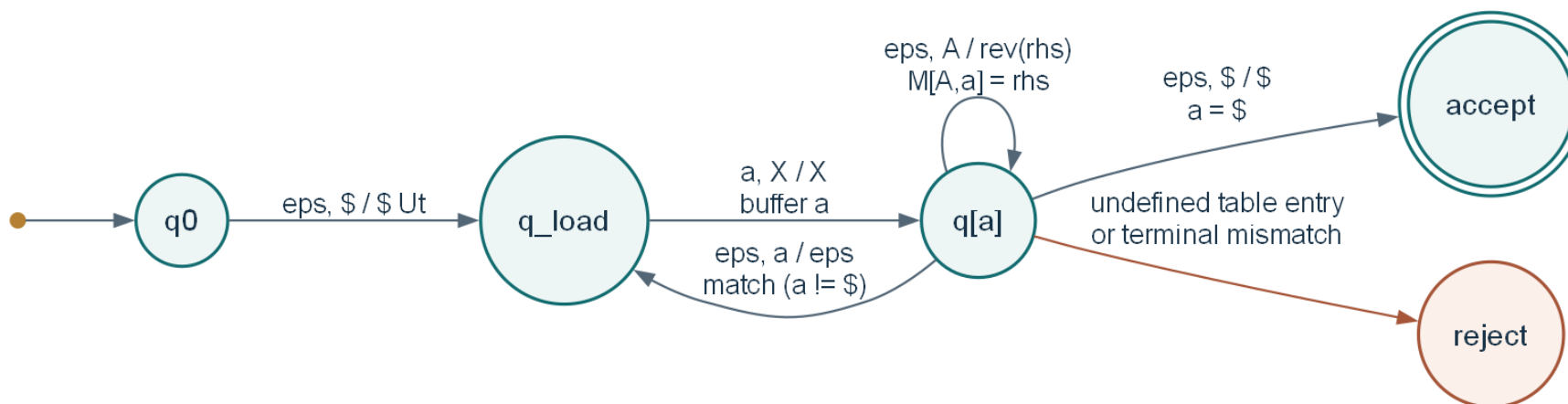
compiler/lexer/tokenizer.py
docs/diagrams/automaton-lexer.puml
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compiler/lexer/lexicon.py

Parser: categories plus a stack

The saved LL(1) table selects each next action.

PARSER / DETERMINISTIC PUSHDOWN CONTROL



EXPAND

Use the unique table entry.

MATCH

Consume one matching category.

FINISH

Accept only at input end.

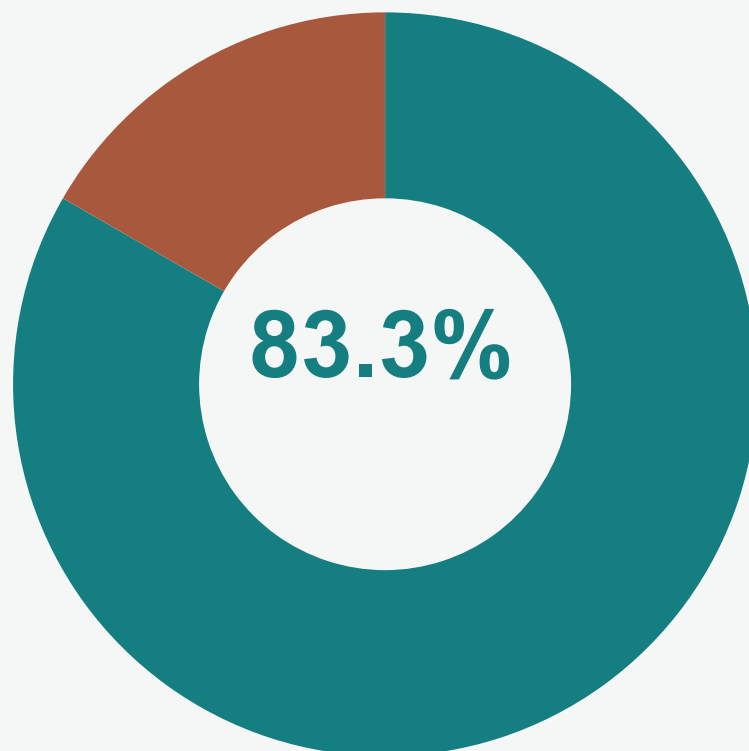
IMPLEMENTATION / EVIDENCE

compiler/parser/analysis.py
docs/diagrams/automaton-parser.puml
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compiler/parser/predictive.py

What the controlled study shows

Twelve supplied statements, 81 token occurrences.



10 ACCEPTED

The category sequences fit the saved grammar.

2 REJECTED

n'ais and alli remain unresolved.

LIMITATION

Grammar match is not linguistic accuracy or proven fieldwork.

IMPLEMENTATION / EVIDENCE

`docs/evidence/final-analysis-20260927.json`

`compiler/parser/yaounde.py`

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`compiler/tests/yaounde_cases.py`

Transparent rules. Visible evidence.

Thank you. Questions?

PRESERVE

The original text

EXPLAIN

The compiler decision

RETAIN

The original result

camfranglais.duckdns.org